A 112-Gb/s Discrete Multitone Wireline Receiver Datapath With Time-Interleaved Time-Based ADC in 5-nm FinFET

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***Abstract*—This article presents a 112-Gb/s discrete multitone (DMT) wireline receiver (RX) datapath with a 50-GS/s, 8-bit, 64-way (8×8) time-interleaved time-based analog-to-digital converter (TI-TBADC) in a 5-nm FinFET. The TBADC converts the voltage input into a time-domain quantity using a ring oscillator (ROSC). Eight-slice TBADCs, driven from the same first-rank interleaver, share the identical injection-locked ROSC (IROSC) for voltage-to-time conversion (VTC). The DMT digital signal processor (DSP) achieves optimal bit and power loading with 63 orthogonal subchannels by employing a 64-way single-stage multi-path delay feedback (MDF) fast Fourier transform (FFT) core. An on-chip sign-sign least mean square (SS-LMS) engine adapts equalizer coefficients to combat channel fluctuation. The RX prototype demonstrates 4E-4 BER when communicating over the channel, exhibiting 18-dB insertion loss (IL) at Nyquist, while consuming 347-mW power and 0.242-mm^2 silicon area.**

***Index Terms*—Bit and power loading, discrete multitone (DMT), injection locking, ring oscillator (ROSC), serial link, time-based analog-to-digital converter (TBADC), wireline transceiver (TRX).**

Received 2 May 2025; revised 10 August 2025; accepted 24 September 2025. Date of publication 14 October 2025; date of current version 30 December 2025. This work was supported in part by the DGIST Research and development Program of the Ministry of Science and ICT under Grant 24-HRHR-02 and Grant 24-KUJoint-09, in part by the Technology Innovation Program funded by the Ministry of Trade, Industry and Energy (MOTIE, Korea) under Grant RS-2024-00410244, in part by the National Research Foundation of Korea (NRF) funded by MSIT under Grant RS-2024-00352784 and Grant RS-2024-00410244, in part by the Institute of Information and Communications Technology Planning and Evaluation (IITP) Grant funded by the Korea Government (MSIT),AI Star Fellowship Support Project (DGIST) under Grant RS-2025-02219277, and in part by the Samsung Future Technology Incubating Program under Grant SRFC-IT2301-01. This article was approved by Associate Editor Hyo Gyuem Rhew. *(Corresponding author: Gain Kim.)*

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Color versions of one or more figures in this article are available at https://doi.org/10.1109/JSSC.2025.3618188.

Digital Object Identifier 10.1109/JSSC.2025.3618188

I. Introduction

THE increasing demands for high-performance computing continue to drive the development of not only optical links but also high-speed electrical interconnects. In hyperscale data center applications, wideband communication over lossy channels requires serializer–deserializer (SerDes) designs with strong equalization and adaptable signal integrity optimization. Recently, the analog-to-digital converter (ADC)-based serial links [1], [2], [3], [4] have addressed these challenges by integrating complex equalization functions into a digital signal processor (DSP) instead of performing in the analog domain.

Conventional ADC-DSP-based SerDes receivers (RXs) [5], [6] employ a continuous-time linear equalizer (CTLE) followed by a time-interleaved (TI) successive approximation register (SAR) ADC. SAR ADCs scale well with technology and exhibit moderate power consumption at high speeds. However, analog components, such as capacitive digital-to-analog converter (CDAC), remain sensitive to linearity and mismatch [7]. As an alternative, time-based (TB) ADCs offer improved scalability by replacing analog sub-blocks with digital circuits [8].

In the digital domain, a long-tap feedforward equalizer (FFE) with a short-tap decision-feedback equalizer (DFE) [9], [10], [11] and optional maximum likelihood sequence detection (MLSD) [12] compensates for severe intersymbol interference (ISI) in four-level pulse amplitude modulation (PAM-4) systems. While the FFE boosts both high-frequency signal components and noise, DFE and MLSD improve the signal-to-noise ratio (SNR) at the decision point. However, the computational complexity of these nonlinear equalizers increases quadratically with the PAM order [13], which hampers the use of higher order PAM symbols for higher bandwidth efficiency. In addition, PAM-4 datapaths are inherently sensitive to spectral notches [14]. While a floating-tap FFE can compensate for reflective postcursors in the time domain, it incurs higher power, area, and noise amplification at the affected frequency bin. In contrast, discrete multitone (DMT) modulation [15], [16], [17] can effectively mitigate the effects of spectral notches by appropriately allocating bits and power across orthogonal subchannels. The bit-loading technique assigns more bits to high-SNR tones and fewer bits to low-SNR tones or even turns off transmission on

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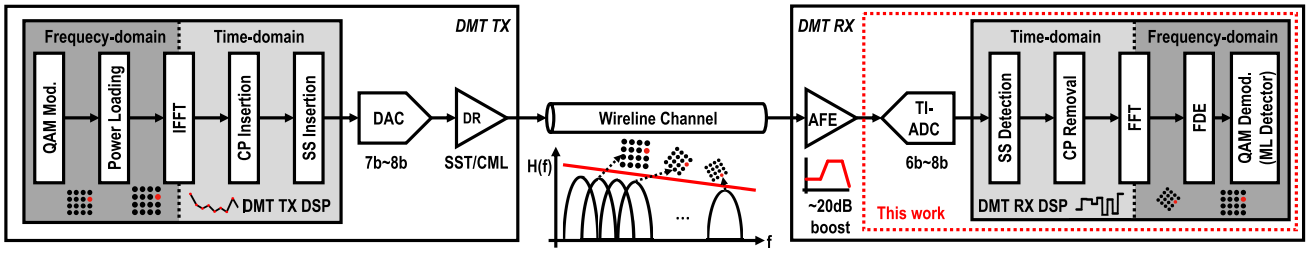


Fig. 1. Overview of a DMT TRX datapath for high-speed serial links.

very-low-SNR tones where a severe spectral notch exists in the frequency domain.

This article presents a 112-Gb/s DMT wireline RX datapath implemented in 5-nm FinFET technology [18]. A 64-way TI-TBADC converts differential input voltages into time-domain signals using a ring oscillator (ROSC)-based time-to-digital converter (TDC) [8]. The TI-TBADC adopts an 8×8 two-stage interleaver architecture, where eight slice TBADCs sharing the same first-rank track-and-hold (T&H) also share an injection-locked ROSC (IROSC), improving area and energy efficiency. Unlike in [8], an IROSC is used to recover the initial state of ROSC for every voltage-to-time conversion (VTC) without an external reset.

The DMT RX DSP utilizes a 64-parallel single-stage 128-tap multi-path delay feedback (MDF) fast Fourier transform (FFT) processor to support subchannel-wise bit and power loading across 63 orthogonal tones. An on-chip sign-sign least mean square (SS-LMS) adaptation engine dynamically adjusts equalizer coefficients in real time to maintain BER performance under varying channel conditions. Frame-level synchronization is supported to detect the DMT symbol starting location for block-based DMT processing.

The remainder of this article is organized as follows. Section II provides a brief review of the DMT transceiver (TRX) datapath for high-speed serial links. Section III presents the overall system architecture of the proposed RX. Section IV details the TI-TBADC design, and Section V describes the DMT DSP architecture. Section VI shows the measurement results, and Section VII concludes this article.

II. DMT TRX Datapath Overview

Fig. 1 provides a brief overview of the DMT TRX datapath for high-speed serial links. Let X_i be a quadrature amplitude modulation (QAM) symbol for the ith subchannel. The power loading scales linearly for each constellation with $\sqrt{P_i}X_i$ to minimize the overall symbol error rate (SER) at the RX, where P_i is the power loading coefficient. The sum of power ($\sum P_i$) is constrained by the peak-to-average-power ratio (PAPR) condition to prevent aggressive clipping. Then, the inverse FFT (IFFT) converts the frequency-domain QAM symbols into the time domain before the cyclic prefix (CP) is inserted [19]. The real-valued transform is used to transmit the time-domain DMT signal over an independent physical lane. The synchronization sequence (SS) is added to find the CP location of DMT symbols on the RX side. Finally, a DAC serializes the parallel digital data with a current-mode logic (CML) or

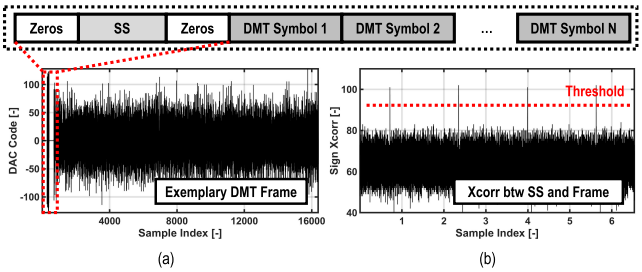


Fig. 2. (a) Exemplary DMT frame and (b) Xcorr between SS and frame.

source-series-terminated (SST) output driver stage. When the CP is sufficiently long to cover ISI, the received symbol Y_i at the RX is represented as follows:

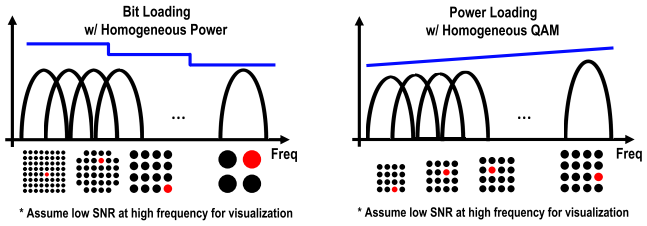
$$Y_i = H_i \sqrt{P_i} X_i + n_i \tag{1}$$

where H_i is the channel response and n_i is the input-referred noise for the ith subchannel. Each QAM symbol is attenuated and phase-rotated by the band-limited wireline channel.

In the RX, an analog front end (AFE), typically including a CTLE and variable gain amplifier (VGA), partially equalizes a lossy channel by boosting high-frequency signal contents by approximately 20 dB [9], [10], [11], prior to quantization and parallelization by a TI-ADC with 6- to 8-bit resolution. The DMT RX DSP detects the SS within the received time-domain DMT frames and removes the CP. An FFT core then converts the time-domain data back into the frequency domain, where a frequency-domain equalizer (FDE) compensates for channel dispersion by restoring both amplitude and phase. Finally, a maximum likelihood (ML) detector demodulates the QAM symbols into binary sequences. This work focuses on prototyping the DMT RX datapath, including TI-ADC and DMT RX DSP, for high-speed serial links.

A single DMT frame consists of an SS placed between zero-padding symbols and DMT data symbols, as shown in Fig. 2(a). Each DMT symbol includes a CP and an IFFT output. The zero-padding prevents ISI between the SS and adjacent DMT symbols. The SS is based on the Zadoff–Chu sequence [20], which exhibits improved autocorrelation and zero cross correlation (Xcorr) properties as compared to random time-domain samples. The RX detects the SS by computing the Xcorr between the known SS and the received time-domain samples and subsequently identifies the CP boundary using the peak location. Since calculating Xcorr requires considerable multipliers and adders, we use SS–Xcorr

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(a) (b)

Fig. 3. (a) Exemplary bit loading with homogeneous power and (b) power loading with homogeneous QAM.

to minimize hardware resources [21] while providing sufficient accuracy as follows:

$$\sum_{i=1}^{L_{\mathrm{ss}}} \mathrm{sign}(\mathrm{ss}[i]) \cdot \mathrm{sign}(y[i+n]) \tag{2}$$

where L_{ss} is the length of the SS and $y[i+n]$ is the received time-domain sample. Fig. 2(b) shows the measured SS-Xcorr result with $L_{\mathrm{ss}} = 128$, demonstrating the effectiveness of SS-Xcorr for high-speed serial links.

Bit and power loading are essential in DMT communication to achieve optimal SER performance for a given channel. Let SNR_i denote the SNR of the ith subchannel. The average bit rate b_i for the ith subchannel is approximated using the SNR gap model [22] as follows:

$$b_i = \log_2\left(1 + \frac{P_i \cdot \mathrm{SNR}_i}{\Gamma}\right) \tag{3}$$

where P_i is the allocated power for the ith subchannel and Γ is the SNR gap, which is inversely proportional to the target SER. According to this equation, higher order QAM symbols are assigned to high-SNR subchannels, while low-SNR subchannels require greater power allocation to maintain a similar SER, as illustrated in Fig. 3. In practice, the total power $\sum_i P_i$ is constrained by the PAPR due to clipping in the data converters. To address this, the loading algorithm can be optimized as follows: 1) maximize $\sum_i b_i$ for a given $\sum_i P_i = \mathrm{const}$ (rate-adaptive) or/and 2) minimize $\sum_i P_i$ for a given $\sum_i b_i = \mathrm{const}$ (margin-adaptive). This work adopts the margin-adaptive loading algorithm to balance the overall SNR under a constant bit rate.

The SNR profile used for loading optimization is estimated from the error vector magnitude (EVM) of equalized QAM-4 pilot symbols as follows:

$$\mathrm{SNR} = -\mathrm{EVM} - \mathrm{PL} \quad [\mathrm{dB}] \tag{4}$$

where PL is the power loading coefficient, and the QAM-4 pilot symbol is assumed to have unit average power. The SNR reflects various noise sources, including thermal noise of the data converter and processing noise of the DSP.

III. Implemented DMT RX With TBADC

Fig. 4 shows an overall block diagram of an implemented 128-tap FFT-based DMT RX datapath with the 64-way TI-TBADC. The RX prototype receives a four-phase quarter-rate clock (C4) and generates the required clocks [e.g., 1/64-rate (C64)] to distribute the interleaver, slice ADCs, retimer, and DSP. The differential input signals (V_{in_p}, V_{in_n}) and clocks are terminated with on-die termination (ODT) using a 100-Ω parallel resistor and are connected to T-coils in series with peaking coils and the electrostatic discharge (ESD) protection circuit. The bi-directional (BIDI) serial interface configures the parameters of ADC/DSP and reads back the on-chip measurement results.

The 64-way TI-TBADC is implemented using a two-stage sampling architecture comprising eight-way bootstrapped T&H switches followed by eight class-AB voltage buffers [23]. Each voltage buffer drives eight slice TBADCs and shares outputs from an eight-stage IROSC. A single IROSC phase updates a 4-bit integer counter for the most significant bits (MSB) before unfolding. The sign bit is detected by a comparator that determines the ramp polarity from the bipolar voltage-to-time converter (BVTC) [8]. When the zero-crossing detector (ZCD) identifies the crossing of differential ramps, the eight IROSC phases are strobed and interpolated by the sense amplifier flip-flop (SAFF) arrays to generate an 8-bit cyclic thermometer code [24], which is then converted to a 4-bit binary value in the DSP. Various pulses for controlling the sign comparator, BVTC, and ZCD are generated on-chip. The 64 parallel outputs from the TBADCs are synchronized using a retimer.

The DMT RX DSP receives 14-bit outputs from the TI-TBADC and converts them into 8-bit binary values after applying offset and gain mismatch calibration in the input logic. Following SS detection via SS-Xcorr, the CP is removed, and the DMT symbol is rearranged to align with the MDF FFT structure. Variable CP lengths are supported to optimize performance across different channels. A 64-way parallel 128-tap real-valued MDF FFT processor transforms the time-domain DMT symbol into frequency-domain QAM symbols. The FDE compensates for channel distortion by scaling and rotating 12-bit input constellations using 8-bit coefficients, which are adaptively updated by the SS-LMS engine to maintain BER under channel variations. The QAM symbols are then demodulated into binary sequences by an ML detector. Additional test circuits, including a digital sampling scope, pseudorandom binary sequence (PRBS) checker, and constellation scanner, are integrated for debugging purposes.

Fig. 5 shows the exemplary output waveform of second-rank T&H and IROSC. The input signal is tracked from the reset state and held for regenerating the sign bit. After detecting the sign bit, the charge is injected to provide the redundancy window against the wrong sign decision with the zero differential input. Then, the input voltage is converted into two time-domain quantities: 1) 4-bit counter output indicating ramp duration multiplied by IROSC frequency and 2) 8-bit cyclic thermometer code representing the state of IROSC at zero crossing. After each conversion, the system is reset to mitigate memory effects.

An 8-bit TBADC output (CNT) is reconstructed by concatenating 4-bit MSB and 4-bit LSB decoded from the 8-bit cyclic thermometer code. Due to the charge injection and initial phase of ISORC, CNT shows a nonzero value (FOLD) for zero

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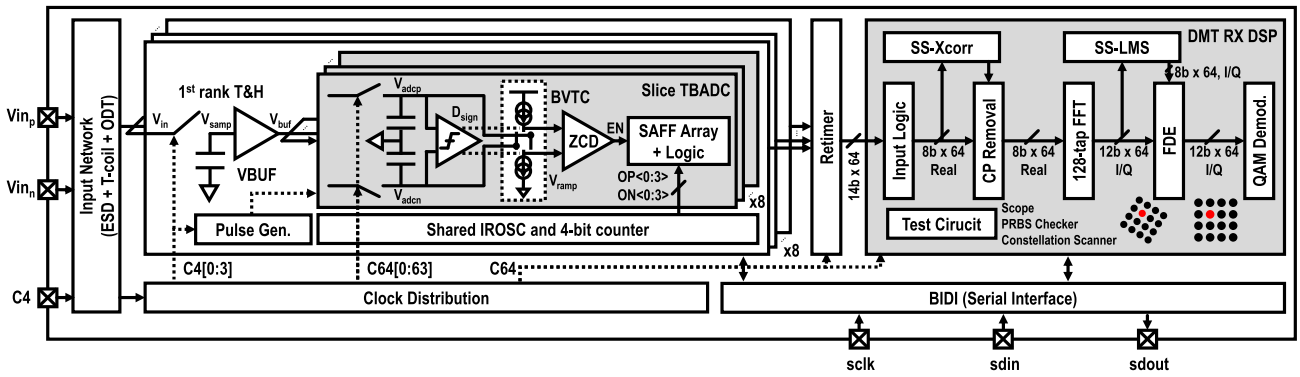


Fig. 4. Overview of a 128-tap FFT-based DMT RX datapath with the 64-way TI-TBADC, where the eight-slice TBADC shares the IROSC.

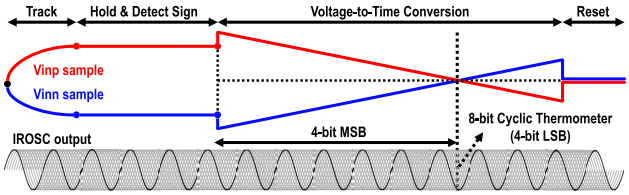


Fig. 5. Exemplary output waveforms of second-rank T&H and IROSC.

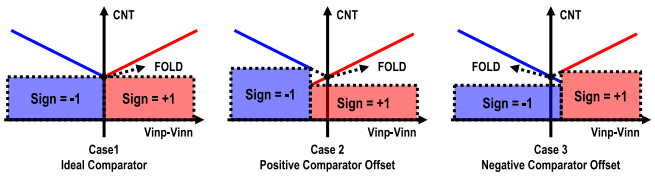


Fig. 6. TBADC output before unfolding for three cases. (a) Ideal comparator, (b) comparator with positive offset, and (c) negative offset.

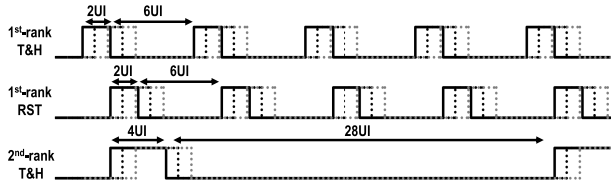


Fig. 7. Interleaver timing diagram.

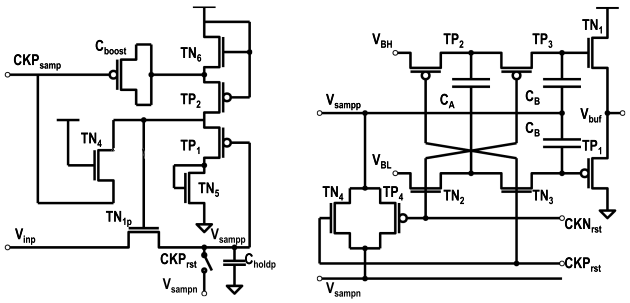


Fig. 8. (a) Schematic of the first-rank bootstrapped T&H switch and (b) class-AB voltage buffer with capacitive biasing network for interleaving.

differential input, as shown in Fig. 6. The comparator offset results in a discontinuous TBADC output response. However, since the conversion gain is identical for both positive and negative signs, the linearly extrapolated CNT intersects FOLD at the y-axis. Therefore, the unfolded 8-bit binary output (D_{out}) is represented as follows:

$$D_{\text{out}} = \begin{cases} +(\text{CNT} - \text{FOLD}) & \text{if Sign} = +1 \\ -(\text{CNT} - \text{FOLD}) & \text{if Sign} = -1. \end{cases} \quad (5)$$

The ADC output codes have the same FOLD values regardless of the sign bit, even with the comparator offset. The LSB decoding and unfolding process for reproducing a binary sample is implemented in the DSP input logic.

Unlike prior work that employs a gated ROSC (GROSC) [8], the proposed TI-TBADC adopts shared IROSCs and counters to eliminate the need for complex gating signals and to amortize the area and power consumption associated with ROSCs. Since the IROSC operates continuously, buffer latency correction logic is required. Additionally, a switched-current (SI) technique is used in the ZCD to mitigate current mismatch between PMOS and NMOS transistors, reducing the number of tunable parameters for gain calibration. Further implementation details are provided in Section IV.

IV. TI-TBADC Details

This section details the TI-TBADC architecture, including the interleaver, shared IROSC, and key subblocks within each slice TBADC, such as the BVTC and ZCD. The MSB buffer latency correction logic is also described.

The interleaver timing diagram, as described in Fig. 7, shows a 25% duty cycle first-rank T&H clock, which is non-overlapping with the second-rank T&H. After second-rank switch tracks input for 4 UI, the reset is raised for 2 UI to short the input signal and charge the bias for the voltage buffer.

Fig. 8(a) shows the schematic of the first-rank bootstrapped T&H switch and the class-AB voltage buffer with a capacitive biasing network. When CKP_{samp} transitions high, TN_6 turns off, and TP_1 is biased through TP_2 by the current from C_{boost}. A PMOS source follower shifts the voltage on the sampling capacitor to the gate of TN_{1p}, maintaining a constant V_{gd}

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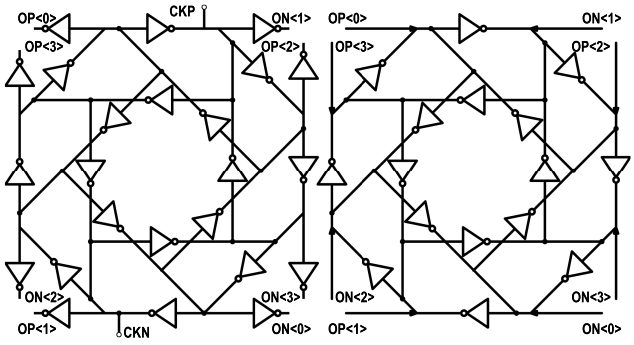


Fig. 9. Schematic of the two-stage IROSC.

to TN_{1p}. As the tracking path has only a single device TP_1, the delay is reduced compared to conventional bootstrapping structures. The TN_5 device limits V_{DS} of TP_1 for protection. When CKP_{samp} transitions low, TN_4 pulls down the gate of TN_{1p}, ensuring a signal-independent turn-off time. As shown in Fig. 8(b), during the reset phase, the voltage buffer shorts TN_4 and TP_4 to reduce the memory effect. The C_B capacitors capture the input common-mode voltage and are connected in parallel with C_A, which is precharged to $V_{BH} - V_{BL}$. In the buffering mode, C_B provides a voltage shift to bias TN_1 and TP_1, while C_A is refreshed to V_{BH} and V_{BL}. The gate voltages of TN_1 and TP_1 are defined by capacitive voltage division, which remains stable across a wide range of input frequencies and minimizes baseline wandering [23].

The IROSC is shared by eight slice TBADCs that use the same first-rank T&H switch. To ensure proper initialization without an external reset, the ROSC frequency is injection-locked to the differential C4 clock (CKP, CKN). The first-stage ROSC receives the C4 input, while the second-stage ROSC receives the eight-phase output of the first stage (OP⟨0:3⟩, ON⟨0:3⟩), as shown in Fig. 9. This two-stage injection-locking scheme enhances the overall locking range. Each stage of ROSC is implemented by the eight-stage inverters, with eight feedback inverters having half the strength of feedforward inverters. This feedback topology enables an even number of output phases while satisfying the Barkhausen criteria. The IROSC phase at the ramp zero crossing is sampled to determine the 4-bit LSB, and a regenerative buffer connected to one of the IROSC phases drives a 4-bit counter that encodes the ramp duration for MSB generation. The regenerative buffer also suppresses glitches that could introduce MSB errors.

Fig. 10 shows the schematics of the sign-bit comparator, BVTC, ZCD, and sense amplifier latch (SAL) in each slice of the TBADC. The sign-bit comparator shown in Fig. 10(a) is implemented using a StrongARM latch with a timeout scheme to ensure BVTC configuration within a fixed evaluation window. When CK_{TO} enables the timeout path, M9 and M10 reinforce regeneration and drive the decision.

During the reset phase, the NMOS current source in Fig. 10(b) charges the SI capacitor to replicate the bias current. When CKP_{ZX} and CKN_{ZX} are enabled, V_{rampp} and V_{rampn} are initialized to GND and VDD, respectively, to suppress memory effects. Based on the sign bit, the BVTC is configured

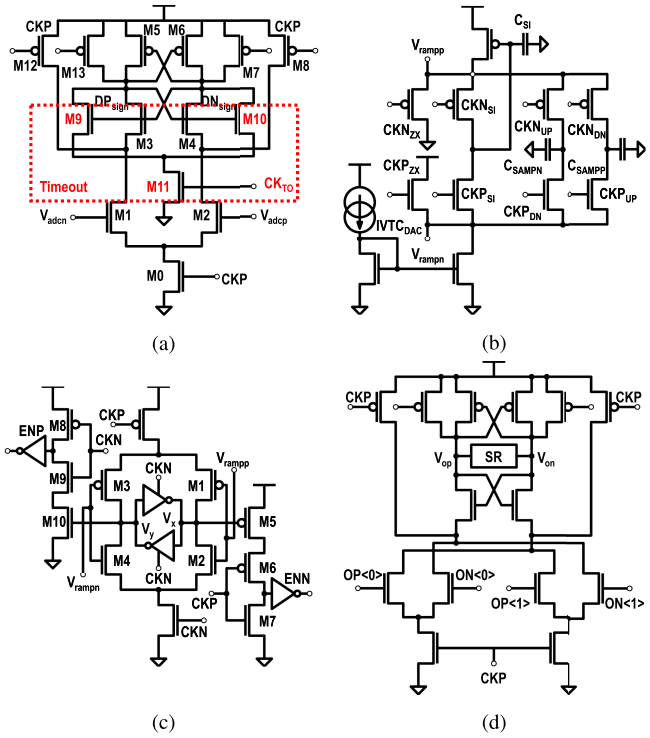


Fig. 10. (a) Schematics of the sign-bit comparator, (b) BVTC, (c) ZCD, and (d) SAL in each slice TBADC.

accordingly. For example, when the sign is positive, CKP_{UP} and CKN_{UP} are enabled, charging V_{rampp} from C_{SAMPN} and discharging V_{rampn} from C_{SAMPP}. Regardless of the sign decision, V_{rampp} and V_{rampn} always generate positive and negative ramps, respectively.

The ZCD in Fig. 10(c) pre-charges ENP and ENN to VDD and GND when CKP is high and CKN is low. Since the BVTC outputs are initialized, V_{rampp} and V_{rampn} begin from GND and VDD, biasing M5 and M10 in the cutoff region. As CKP and CKN transition low and high, respectively, the cross-coupled inverters are triggered once V_x and V_y reach the trip point during ramp crossing. This activates M5 and M10, charging ENP and discharging ENN. The positive feedback in cross-coupled inverters prevents glitches at the output node. The eight-parallel SAL in Fig. 10(d) captures the IROSC phase at the ramp-crossing instant detected by the ZCD. Phase interpolation is used to resolve time steps further and generate an 8-bit cyclic thermometer code for the 4-bit LSB. The SAL is chosen to minimize static power.

Fig. 11 shows the asynchronous counter decoder and buffer latency correction logic. The 4-bit ripple counter consists of two cascaded divide-by-four stages and is driven by a voltage buffer connected to one of the IROSC phases. The SAFF strobes the 4-bit counter output with respect to CK SAFF asynchronously. To mitigate metastability, both the direct counter output A[3:0] and its half-cycle delayed version B[3:0] are sampled. The asynchronous counter decoder selects either A or B based on the sampled IROSC phase and converts the selected value into binary using XOR gates [7]. Due to the latency of the voltage buffer driving the counter, the LSB may be sampled earlier than the MSB, potentially causing

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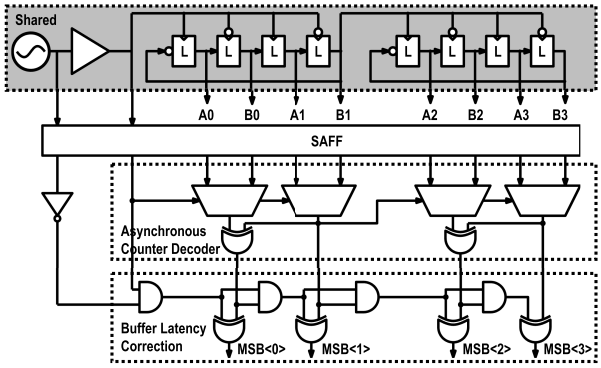


Fig. 11. Asynchronous counter decoding and buffer latency correction logic.

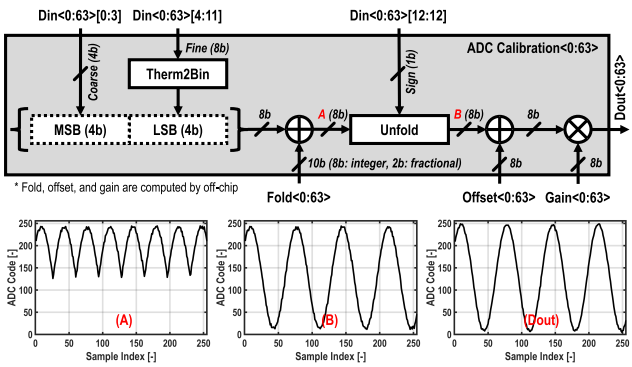


Fig. 12. DSP input logic layer for TI TB-ADC.

a 16-code error during the rollover from 15 to 0. To address this, the buffer latency correction logic compares the sampled IROSC with the sampled buffer output and increments the MSB by one when LSB rollover is detected.

V. DMT RX DSP Details

This section presents the implementation of the DMT RX DSP datapath, including the TB-ADC interface, synchronization, CP removal, FFT, FDE, and SS-LMS adaptation engine. DSP routing congestion and cell placement strategies are also discussed.

Fig. 12 shows the DSP input logic that converts the 13-bit output of the TI TB-ADC into an 8-bit signed binary code. The input consists of a 4-bit MSB, an 8-bit cyclic thermometer code, and a 1-bit sign. The thermometer code is decoded into a 4-bit binary value, with bubble codes suppressed to zero. A 10-bit folding point is added to the decoded binary value, and unfolding is performed based on the sign bit. The offset and gain mismatch between slice ADCs is corrected in the digital domain. The folding point, offset, and gain calibration parameters are computed off-chip.

The 64-parallel, 128-input SS-Xcorr detects the CP location by computing the Xcorr between the ADC-calibrated output and a predefined SS, such as the Zadoff–Chu sequence with ideal autocorrelation properties. To reduce DSP area and power consumption, arithmetic multiplications are approximated by sign-sign XNOR operations [21], as illustrated in Fig. 13. A synchronization flag is asserted when the SS-Xcorr output

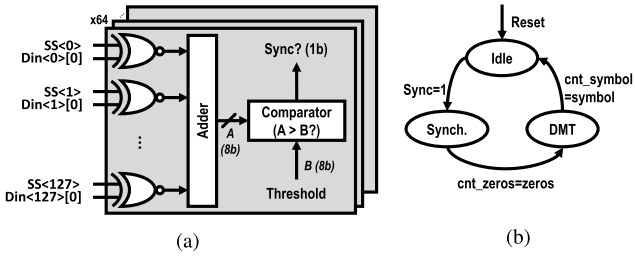


Fig. 13. (a) Block diagram of SS-Xcorr engine and (b) FSM for synchronization.

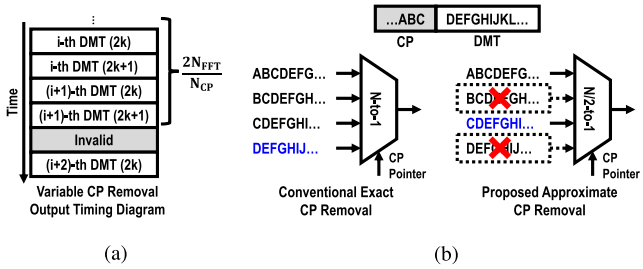


Fig. 14. (a) Timing diagram of variable CP removal with serial commutation and (b) comparison between conventional and proposed CP removal.

exceeds a programmable threshold. The DSP operates in three synchronization states: Idle, Synch, and DMT. Upon SS detection, the state transitions to Synch and remains there until the number of detected zeros matches the predefined count (cnt_zeros). The FSM then enters the DMT state, during which it counts the number of received DMT symbols. Once the symbol count reaches cnt_symbol, the FSM returns to the idle state.

The primary function of the CP removal block is to commutate time-domain DMT samples for proper MDF FFT processing while supporting variable-length CP elimination. As shown in Fig. 14, each N_{FFT}-point DMT symbol is divided into even- and odd-indexed elements and serially pipelined into the MDF FFT core. A validity flag is asserted every $2 \times N_{\mathrm{FFT}}/N_{\mathrm{CP}}$ C64 clock cycles, reducing the effective throughput proportional to the CP overhead. While longer CP lengths help suppress residual ISI, they degrade data rate efficiency given Baudrate. Therefore, the CP length must be optimized based on the channel characteristics. Supporting various CP lengths requires wide N-to-1 multiplexer (MUX) arrays, causing significant routing congestion. N is chosen to be 256, which is twice that of N_{FFT}. To mitigate this, an approximate CP removal scheme is implemented by reducing the MUX width from N-to-1 to $N/2$-to-1. This approximation introduces at most a one-sample CP location offset, which is acceptable when a sufficiently long CP is used to absorb residual ISI.

A large number of subchannels benefit DMT systems by distributing the energy of residual ISI across the symbols and mitigating data rate degradation due to CP overhead. However, implementing a high-throughput, large-tap FFT remains challenging under tight power and area constraints [21]. In this work, a 64-parallel 128-tap real-valued MDF FFT is implemented by serially folding a 64-tap FFT twice, as shown

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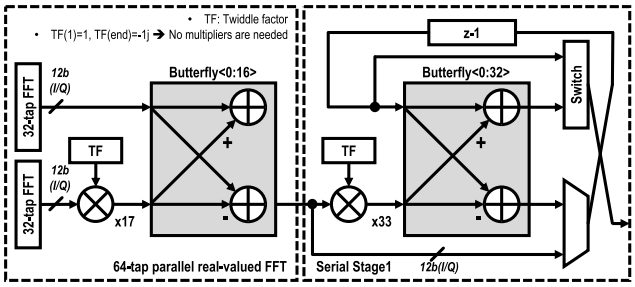
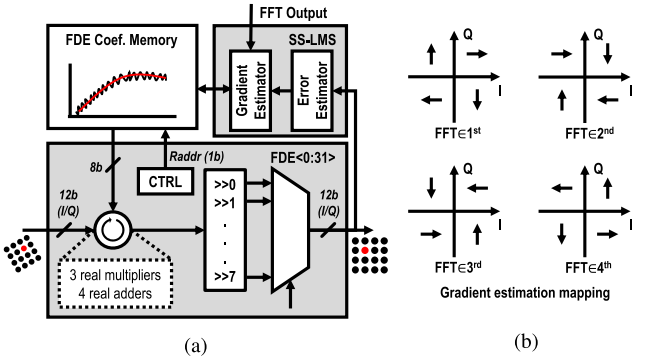


Fig. 15. Block diagram of 64-parallel 128-tap real-valued MDF FFT.



(a) (b)

Fig. 16. (a) Block diagram of adaptive FDE with SS-LMS engine and (b) gradient estimation mapping.

in Fig. 15. The 64-tap FFT adopts a decimation-in-time (DIT) architecture composed of two 32-point parallel FFT stages and butterfly networks as follows:

$$Y_N[m] = \sum_{k=1}^{N} y[k] W_N^{mk} \tag{6}$$

$$= W_N^{-m} \sum_{k=1}^{N/2} y[2k-1] W_{N/2}^{mk} + \sum_{k=1}^{N/2} y[2k] W_{N/2}^{mk} \tag{7}$$

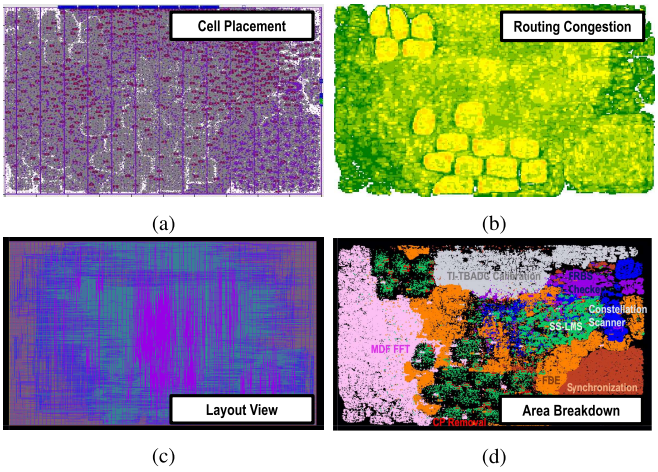
where $y[k]$ denotes the input time-domain samples and $Y_N[m]$ represents the output frequency-domain samples of an N-point FFT. The twiddle factor (TF) is defined as

$$W_N = \exp\left(-j\frac{2\pi}{N}\right) \tag{8}$$

and each W_N^m is quantized into 8-bit fixed-point format, consisting of a 1-bit sign and a 7-bit fractional part. The number of complex multiplications for TF computation increases linearly with the FFT size. To reduce area, all TF multiplications are performed using constant coefficients, eliminating the need for full-complex multipliers and halving the multiplication logic. Multiplications with trivial TFs, such as ± 1 and $\pm j$, are implemented using sign inversion and wiring manipulation. Due to the conjugate symmetry of real-valued FFTs, only half of the butterfly structure is required. In the second stage, (7) is computed sequentially for $N = 128$. The even-indexed outputs are delayed by one C64 clock cycle and combined with TF-multiplied odd-indexed outputs at the butterfly nodes. Finally, the even and odd subchannel results are interleaved using switch logic. The available area was insufficient to implement a 256-tap FFT in the 5-nm technology.

The datapath precision (i.e., 12 bit in Fig. 15) is optimized based on the input statistics of DMT, assuming a 12-dB PAPR at the RX. Pipelined registers are inserted only at the butterfly output nodes of each stage in the parallel 64-tap FFT (except for 2-tap FFT), indicating that both the complex multiplication and butterfly operations are performed within a single C64 clock cycle. In the serial stage, pipeline registers are placed at the delay elements and switch logic. As a result, the 64-parallel 128-tap MDF FFT exhibits a total latency of seven C64 clock cycles—five cycles for the 64-tap parallel FFT and two cycles for the serial stage.

The 32-parallel FDE expands and rotates the received 12-bit constellations through complex multiplication with 8-bit FDE coefficients, as shown in Fig. 16(a). Each complex multiplier



(a) (b) (c) (d)

Fig. 17. (a) DSP cell placement, (b) routing congestion map, (c) layout, and (d) area breakdown.

is implemented using three real multipliers and four real adders to minimize hardware cost. Due to the interleaving of even and odd subchannel outputs from the MDF FFT, the FDE controller generates a 1-bit read address (Raddr) to load the corresponding coefficients from the coefficient memory. The coefficients are stored in fixed-point format with variable binary points, enabling dynamic range optimization and minimizing quantization noise. Thus, the FDE output $\hat{X}_i$ is represented as

$$\hat{X}_i = \frac{\lfloor 2^{M_i} \cdot F_i \rceil \cdot Y_i}{2^{M_i}} \tag{9}$$

where M_i denotes the binary point position, F_i is the FDE coefficient, Y_i is the FFT output for the ith subchannel, and $\lfloor \cdot \rceil$ denotes quantization. The division by 2^{M_i} is implemented using a shift-right operation, where M_i is computed off-chip and used to configure the MUX selection. Saturation logic is included at the output stage to prevent overflow. This format is particularly effective for low-frequency subchannels with small-magnitude coefficients. The FDE block includes pipeline registers before the complex multiplication and after the MUX operation, resulting in a latency of two C64 clock cycles.

The FDE coefficients can be adapted by the SS-LMS engine to maintain BER under channel variation [22]. The error vector is estimated by directly subtracting the equalized

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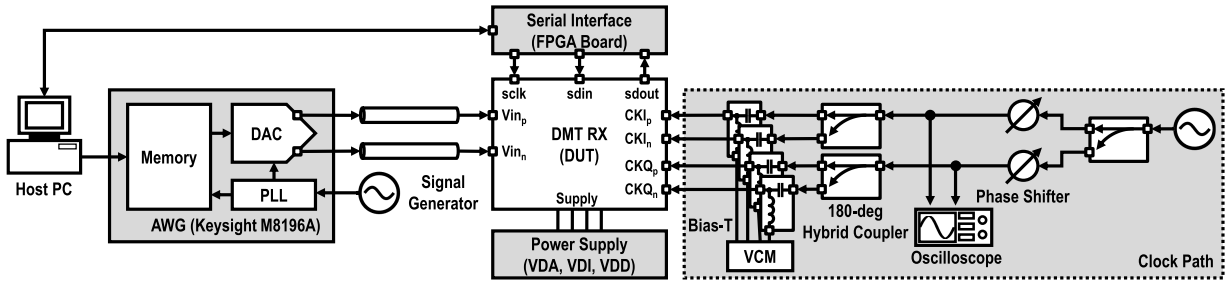


Fig. 18. Measurement setup.

constellation from the symbol decision output. Then, the SS-LMS engine updates the coefficients by adding a gradient in the negative direction with a fixed step size μ. As shown in Fig. 16(b), the gradient direction is mapped using a four-input lookup table (LUT), which encodes the signs of the error vector and the corresponding FFT output. The FFT output is pipelined by two C64 clock cycles to compensate for the FDE processing latency. Thus, the approximated gradient of the FDE coefficient is given by

$$\nabla F_i \simeq \mu \cdot (\alpha + j\beta) \tag{10}$$

where α and β are either ± 1, depending on the polarity of the FFT and FDE outputs. The FDE coefficients are pipelined by one C64 clock cycle after gradient update.

Fig. 17 shows the physical implementation results of the RX DSP, including cell placement, routing congestion, layout view, and area breakdown, synthesized for a 1.25-GHz target clock. The reported cell density is 36.24% with 632.1k standard cells. Significant routing congestion is observed around the memory blocks used for CP removal and FDE coefficient storage. The layout shown in Fig. 17(d) highlights the area contribution of each major functional block.

VI. Measurement Results

Fig. 18 shows the measurement setup used to characterize the DMT RX test chip mounted on a PCB. The M8196A arbitrary waveform generator (AWG) transmits pre-computed sequences stored in its memory via an 8-bit DAC. The DAC operates at a rate of 32 times higher than the external clock frequency and 1.5 times the ADC sampling rate. Sequences, such as SS or DMT frames, are generated by a host PC and loaded into the DAC memory. Differential input signals are dc-coupled to the input pads ($V_{\text{in},p}$, $V_{\text{in},n}$), with 350-mV common-mode voltage and 2-V_{ppd} output swing programmed by the AWG. Four-phase clocks are derived from a single signal generator and split using cascaded 180° hybrid couplers and mechanical phase shifters for I/Q generation. IQ mismatch is monitored via an oscilloscope, and interconnect delay is adjusted using phase shifters. Clock signals are ac-coupled to the pads, while bias-T circuits provide common-mode levels. The AWG and RX clocks are synchronized with each other to minimize sampling frequency offset. A host interface board configures RX parameters and retrieves measurement data via a three-wire serial interface. All control and analysis tasks are managed by a PC. The DMT RX test chip is powered by four supplies: VDA for slice TBADCs, VDI for interleaver, VDD for DSP, and VDC for the clock. A high-resistance analog MUX (AMUX) measures the supply inside the chip after the IR drop.

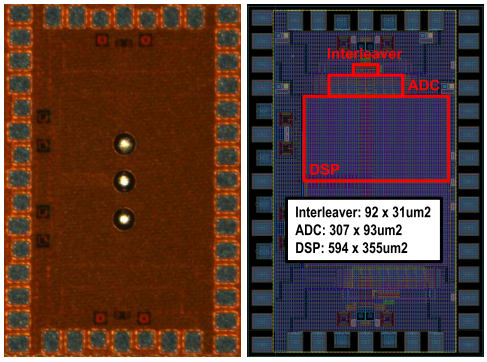
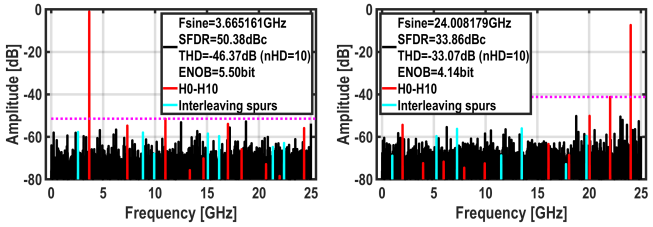


Fig. 19. Die photograph and layout details.



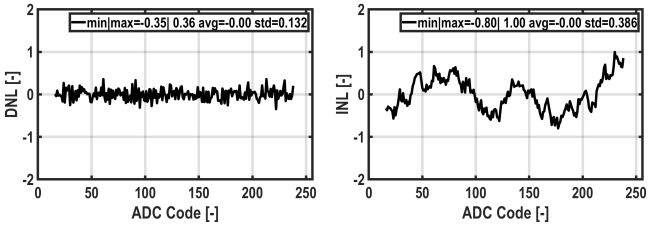
(a) (b)

Fig. 20. ADC output spectrums corresponding to (a) low-frequency and (b) high-frequency input at a 50-GS/s sampling rate.

Fig. 19 shows the die photograph and layout of the DMT RX implemented in 5-nm FinFET. The areas of the clock RX, interleaver, ADC, and DSP are 46×21 μm^2, 92×31 μm^2, 307×93 μm^2, and 594×355 μm^2, respectively, resulting in a total active area of 0.242 mm^2. The synthesized DSP contains 632.1k standard cells with a cell density of 36.24%.

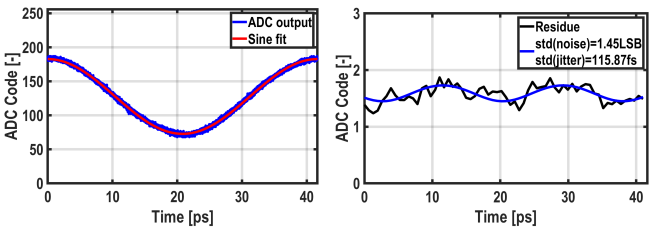
For ADC characterization, a PSG vector signal generator is used to generate sinusoidal inputs, and an on-DSP digital sampling scope collects 16k samples after binary format conversion logic. Fig. 20 shows the ADC output spectra for Fig. 20(a) low-frequency and Fig. 20(b) high-frequency inputs at a 50-GS/s sampling rate. Calibration parameters for FOLD, offset, gain, duty cycle, and skew are obtained at the chip start-up. Due to the 18-dB insertion loss (IL), the high-frequency signal is measured with a 7.3-dB loss.

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(a) (b)

Fig. 21. (a) DNL and (b) INL curves from the sinusoidal histogram test.



(a) (b)

Fig. 22. (a) Modulo-time plot and (b) cosine fitting for sensitivity and jitter extraction.

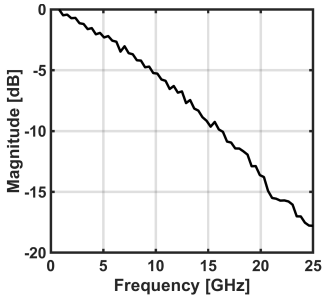
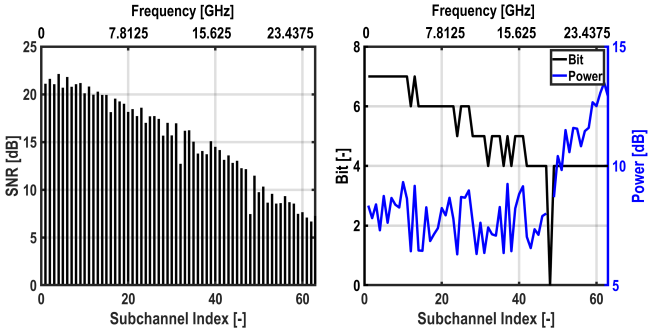


Fig. 23. Measured channel frequency response.



(a) (b)

Fig. 24. (a) Channel SNR profile and (b) bit and power loading profile.

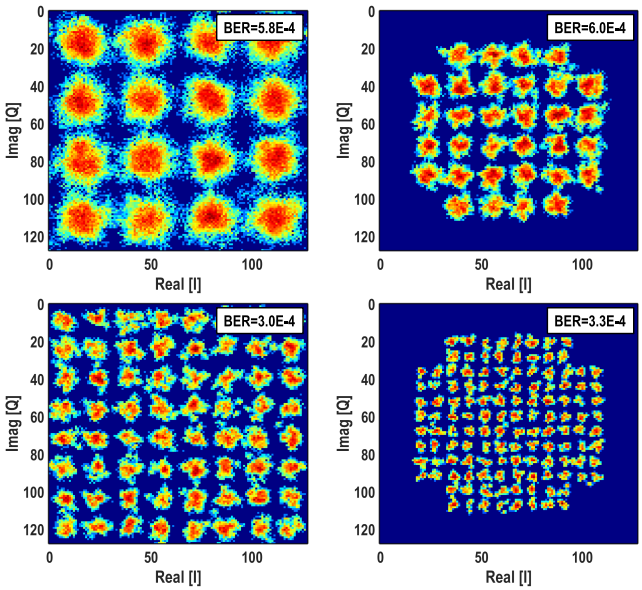


Fig. 25. Scanned QAM-16/32/64/128 constellation diagrams after FDE with corresponding BER.

Fig. 21 shows the integral nonlinearity (INL) and differential nonlinearity (DNL) curves measured from a sinusoidal histogram test using a 3.67-GHz sine input. The DNL remains between −0.35 and +0.36 LSB, and the INL stays within −0.80 to +1.00 LSB, with no missing codes observed. Fig. 22(a) illustrates the modulo-time plot and Fig. 22(b) corresponding cosine fitting used for jitter extraction with a 24.01-GHz sine input. For an ideal 7-bit quantizer with 7.3-dB back-off, the extracted jitter of 115.87 fs corresponds to an effective number of bits (ENOB) of 5.4 bits. Thus, the measured SNR near Nyquist remains limited by random noise rather than jitter.

Fig. 23 shows the measured channel frequency response, showing 18 dB IL at 25-GHz Nyquist frequency. The loss includes the effect of the chip front end, PCB traces, and connectors. The SNR shown in Fig. 24 is measured based on the EVM of the equalized QAM-4 pilot symbols. The loading profile is computed using a margin-adaptive Levin–Campello (LC) algorithm [25], targeting an SER of 2.31E−3 per dimension. The 48th subchannel is excluded from transmission due to a strong quarter-rate interleaving spur from the AWG, which severely distorts the constellation, and the linear FDE is used to recover the remaining QAM symbols. The measured PAPR is 12.1 dB, and the achieved average bit rate is 5.0625 b/Hz, taking account of CP overhead.

The constellation diagrams captured by the on-chip scanner, shown in Fig. 25, exhibit EVMs of −16.0, −18.9, −22.5, and −25.3 dB for QAM-128, QAM-64, QAM-32, and QAM-16, respectively. An average BER of 4.4E−4 is achieved over a channel with 18 dB IL at a 25-GHz Nyquist frequency.

Fig. 26 shows the measured power dissipation breakdown for Fig. 26(a) analog and Fig. 26(b) digital domains. The total RX power consumption is 347 mW, corresponding to an energy efficiency of 3.09 pJ/b at a 112-Gb/s data rate. The voltages of the supplies are measured on-chip. The analog circuits consume 117.6 mW at 50 GS/s with 0.70-V VDI and 0.65-V VDA, while the DSP consumes 229.6 mW at a 781.25-MHz clock frequency with 0.625-V VDD. The digital power is dominated by the 128-tap FFT (34 %).

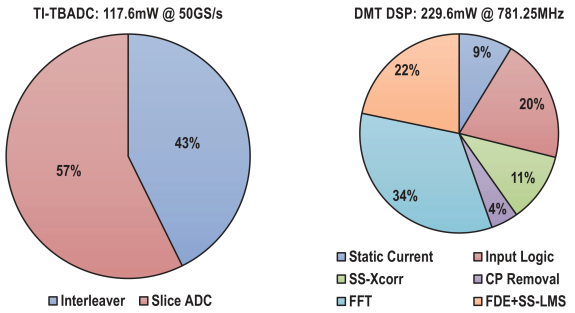
Table I summarizes the features of the implemented DMT RX prototype and compares them with prior high-speed wireline RX designs with similar data rate or modulation scheme. The test chip demonstrates the first DMT RX achieving

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TABLE I
COMPARISON TO PRIOR ARTS ON HIGH-SPEED WIRELINE RX

	This work	[15]	[3]	[5]	[6]
Technology	5nm	14nm	7nm	5nm	5nm
Data Rate (Gbps)	112.5	56	112	112.5	112.5
Modulation	DMT	DMT	PAM-4	PAM-4	PAM-4
Supply Voltage (V)	0.70/0.65/0.625	0.8/0.7/0.7/0.75	1.0/0.68	0.75/1.0	0.75
Synchronization	On-chip	Off-chip	-	-	-
RX Architecture	AFE: N.A.	AFE: N.A.	AFE: VGA + CTLE	AFE: VGA + CTLE	AFE: ATT* + VGA + CTLE
	ADC: 64-way TI 8b TBADC	ADC: 48-way TI 10b SARADC	ADC: 64-way TI 7b SARADC	ADC: 64-way TI 7b SARADC	ADC: 56-way TI 7b SARADC
	DSP: 128-tap FFT + FDE	DSP: 32-tap FFT + FDE	DSP: 5-tap FFE + 9-tap SB-DFE	DSP: 22+8-tap FFE +1-tap DFE	DSP: 24+8-tap FFE +1-tap DFE
BER @ Loss	4e-4 @ 18 dB	2e-4 @ 20 dB	2e-6 @ 45 dB	1e-9 @ 40 dB	7e-6 @ 48 dB
Power (DSP) (mW)	347 (230)	161 (68)	N.A. (175)	504 (202)*****	521 (173)*****
Energy Efficiency (pJ/b)	3.14	2.88	1.56***	4.48*****	4.63*****
Area (mm^2)	0.242	0.26	0.023**	0.49****	0.461****

* ATT: Analog attenuator, ** RX DSP cell area, *** RX DSP power, **** TRX area, ***** TRX power



(a) (b)

Fig. 26. Power dissipation breakdowns for (a) analog and (b) digital.

a >100-Gb/s data rate with a digital-friendly TI-TBADC architecture, while maintaining robust performance in the presence of a deep notch in the end-to-end channel SNR profile.

VII. CONCLUSION

This article presents the 112-Gb/s DMT RX datapath integrated with the TI-TBADC in 5-nm FinFET. The IROSC that converts time-domain quantities into digital codes is shared among eight slice TBADCs for area and power saving. Optimal bit and power loading across 63 orthogonal subchannels is achieved using a 64-parallel, single-stage MDF 128-tap FFT. The test chip occupies 0.242 mm^2 and consumes 347 mW while achieving a BER of 4E−4 over a channel exhibiting a notch at bin 48- and 18-dB IL at Nyquist.

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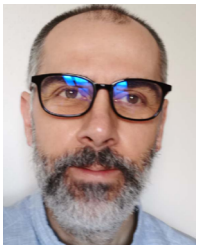
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